HUNDRED DAYS OF CODE
DAY 29

QUESTION:

Write a C program to perform the following menu-driven operations:

1. Count even numbers, odd numbers, positive, negative, and zero elements in an array.
2. Search for an element in an array using linear search.
3. Merge 2 arrays.
4. Exit

SOLUTION:

1. Aim

To write a menu-driven C program using a switch statement to perform various operations on arrays, including counting even, odd, positive, negative and zero elements, searching for an element using linear search, and merging two arrays.

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2. Algorithm

Main Algorithm

1. Start the program.
2. Display the following menu:
    - Count even, odd, positive, negative and zero elements.
    - Search for an element using linear search.
    - Merge two arrays.
    - Exit.
3. Accept the user's choice.
4. Use a switch statement to perform the selected operation.
5. Repeat the menu until the user selects **Exit**.
6. Stop.

Operation 1: Count Array Elements

1. Input the number of elements n.
2. Input n elements into the array.

HUNDRED DAYS OF CODE
DAY 29

3. Initialize even, odd, positive, negative, and zero counters to 0.
4. Traverse the array using a loop.
5. If an element is divisible by 2, increment even; otherwise, increment odd.
6. If an element is greater than 0, increment positive.
7. If an element is less than 0, increment negative.
8. Otherwise, increment zero.
9. Display all five counts.

Note: 0 is an **even number**, so it is included in the even count. However, it is neither positive nor negative.

Operation 2: Linear Search

1. Input the number of elements.
2. Input the array elements.
3. Input the element to be searched.
4. Traverse the array from the first element to the last.
5. Compare each array element with the search element.
6. If a match is found, display its position and stop searching.
7. If no match is found after traversing the entire array, display "Element not found".

Operation 3: Merge Two Arrays

1. Input the size and elements of the first array.
2. Input the size and elements of the second array.
3. Copy all elements of the first array into the merged array.
4. Copy all elements of the second array after the elements of the first array.
5. Display the merged array.

C PROGRAM:

```
#include <stdio.h>


#define MAX 100
```

HUNDRED DAYS OF CODE
DAY 29

```
int main()
{
   int choice;

   do
   {
      printf("\n----- ARRAY OPERATIONS MENU -----\n");
      printf("1. Count even, odd, positive, negative and zero elements\n");
      printf("2. Search an element using Linear Search\n");
      printf("3. Merge two arrays\n");
      printf("4. Exit\n");

      printf("Enter your choice: ");
      scanf("%d", &choice);

      switch(choice)
      {
         case 1:
         {
            int arr[MAX], n, i;
            int even = 0, odd = 0;
            int positive = 0, negative = 0, zero = 0;

            printf("Enter number of elements: ");
            scanf("%d", &n);

            printf("Enter %d elements:\n", n);
```

HUNDRED DAYS OF CODE
DAY 29

```
for(i = 0; i < n; i++)
{
   scanf("%d", &arr[i]);
}


for(i = 0; i < n; i++)
{
   // Count even and odd numbers
   if(arr[i] % 2 == 0)
      even++;
   else
      odd++;

   // Count positive, negative and zero
   if(arr[i] > 0)
      positive++;
   else if(arr[i] < 0)
      negative++;
   else
      zero++;
}

printf("\nEven elements     = %d\n", even);
printf("Odd elements      = %d\n", odd);
printf("Positive elements = %d\n", positive);
printf("Negative elements = %d\n", negative);
```

```
    printf("Zero elements     = %d\n", zero);


    break;
}


case 2:
{
    int arr[MAX], n, i, key;
    int found = 0;


    printf("Enter number of elements: ");
    scanf("%d", &n);


    printf("Enter %d elements:\n", n);


    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }


    printf("Enter element to search: ");
    scanf("%d", &key);


    // Linear Search
    for(i = 0; i < n; i++)
    {
        if(arr[i] == key)
```

```
      {
         printf("Element %d found at position %d.\n",
              key, i + 1);
         found = 1;
         break;
      }
   }

   if(found == 0)
   {
      printf("Element not found in the array.\n");
   }

   break;
}

case 3:
{
   int arr1[MAX], arr2[MAX], merged[2 * MAX];
   int n1, n2, i;

   printf("Enter size of first array: ");
   scanf("%d", &n1);

   printf("Enter elements of first array:\n");

   for(i = 0; i < n1; i++)
```

HUNDRED DAYS OF CODE
DAY 29

```
{
   scanf("%d", &arr1[i]);
}

printf("Enter size of second array: ");
scanf("%d", &n2);

printf("Enter elements of second array:\n");

for(i = 0; i < n2; i++)
{
   scanf("%d", &arr2[i]);
}

// Copy first array
for(i = 0; i < n1; i++)
{
   merged[i] = arr1[i];
}

// Copy second array
for(i = 0; i < n2; i++)
{
   merged[n1 + i] = arr2[i];
}

printf("Merged array:\n");
```

HUNDRED DAYS OF CODE
DAY 29

```
            for(i = 0; i < n1 + n2; i++)
            {
                printf("%d ", merged[i]);
            }

            printf("\n");

            break;
        }

        case 4:
            printf("Program exited successfully.\n");
            break;

        default:
            printf("Invalid choice! Please try again.\n");
    }

  } while(choice != 4);

  return 0;
}
```

TEST CASE TABLE:

HUNDRED DAYS OF CODE
DAY 29

Test Case	Choice	Input	Expected Result
1	1	5: 2 -3 0 7 -4	Even = 3, Odd = 2, Positive = 2, Negative = 2, Zero = 1
2	1	4: 2 4 6 8	Even = 4, Odd = 0, Positive = 4
3	2	5: 10 20 30 40 50, Search = 30	Element found at position 3
4	2	4: 5 10 15 20, Search = 7	Element not found
5	3	1 2 3 and 4 5 6	Merged = 1 2 3 4 5 6
6	4	—	Program exits successfully
7	7	—	Displays invalid choice message

HUNDRED DAYS OF CODE
DAY 29

COMPILATION AND EXECUTION:

```
┌──(kali㉿kali)-[~/hdoc]
└─$ vi h29.c

┌──(kali㉿kali)-[~/hdoc]
└─$ cc h29.c

┌──(kali㉿kali)-[~/hdoc]
└─$ ./a.out

──── ARRAY OPERATIONS MENU ────
1. Count even, odd, positive, negative and zero elements
2. Search an element using Linear Search
3. Merge two arrays
4. Exit
Enter your choice: 1
Enter number of elements: 5
Enter 5 elements:
2 -3 0 7 -4

Even elements     = 3
Odd elements      = 2
Positive elements = 2
Negative elements = 2
Zero elements     = 1
```

```
──── ARRAY OPERATIONS MENU ────
1. Count even, odd, positive, negative and zero elements
2. Search an element using Linear Search
3. Merge two arrays
4. Exit
Enter your choice: 2
Enter number of elements: 5
Enter 5 elements:
10 20 30 40 50
Enter element to search: 30
Element 30 found at position 3.

──── ARRAY OPERATIONS MENU ────
1. Count even, odd, positive, negative and zero elements
2. Search an element using Linear Search
3. Merge two arrays
4. Exit
Enter your choice: 3
Enter size of first array: 3
Enter elements of first array:
1 2 3
Enter size of second array: 4
Enter elements of second array:
4 5 6 7
Merged array:
1 2 3 4 5 6 7

──── ARRAY OPERATIONS MENU ────
1. Count even, odd, positive, negative and zero elements
2. Search an element using Linear Search
3. Merge two arrays
4. Exit
```